

A Concise Guide for Conductors of Amateur Orchestras—Second Edition

Wang Leda

Date: July 19, 2026

Contents

1	Preface to the Second Edition	2
2	Preface	3
3	The Composition of an Amateur Symphony Orchestra	5
4	Who Is Suited to Conduct an Amateur Symphony Orchestra?	6
5	How Does One Learn to Conduct? Technical Aspects	7
5.1	Instrumentation	8
5.2	Music Theory	8
5.3	Ear Training	9
5.4	Score Reading and Analysis; Evaluating Repertoire	10
5.5	Music History, Major Composers, and Important Repertoire	11
5.6	Basic Conducting Technique: To Beat Time or Not to Beat Time?	13
6	How to Lead a Rehearsal: Communication	14
7	Orchestral Training: Rehearsal Technique	16
8	Orchestral Training: Psychology	18
9	Instrumentation for Amateur Orchestras	21
9.1	General Considerations for String Technique	22
9.2	General Considerations for Woodwind Technique	22
9.3	General Considerations for Brass Technique	23
9.4	General Considerations for Percussion Technique	23

1 Preface to the Second Edition

The second edition of this guide was expanded after conversations with students with whom the author had worked in the Student Orchestra of his college. Over the past two years, while continuing his studies, the author has explored different musical genres and styles and conducted several thought experiments on specific questions of musical performance. During the same period, the orchestra grew to an unprecedented size, its instrumentation became increasingly complete, and students showed greater enthusiasm for practicing. Under these changed circumstances, orchestral rehearsals placed higher demands on the conductor's own abilities. In view of the problems currently found in rehearsals, the author decided to revise part of this document, which had originally been written rather hastily. Although it is called a "guide," repeated revisions have left it somewhat disorderly; it is closer to a collection of experience and a practical memorandum. The hope is that it can continue to help those who now or in the future conduct rehearsals for orchestral ensembles.

The author and his colleagues have repeatedly conducted thought experiments (Gedankenexperimente) and, after much consideration, have concluded that a guiding document of this kind is still necessary. In every historical period, someone will say things such as "a professional music director should be hired to handle the orchestra's affairs" or "students cannot effectively lead a student orchestra." From the standpoint of organizational operation alone, such claims are not without reason. Yet a large part of what makes a student orchestra uniquely valuable is precisely its greatest possible degree of independence and self-governance. The value of a student orchestra is not reflected only in the quality of its final performance. It is also a form of practice in which students organize themselves, learn together, and assume artistic responsibility. When students conduct, the choice of repertoire, rehearsal methods, artistic judgments, and the orchestra's future direction are discussed and decided mainly by the participating students. Such a process may be less efficient than professional management and will inevitably involve mistaken judgments, insufficient technique, and repeated experimentation. Yet it is precisely through these imperfect practices that members genuinely learn to listen to one another, express opinions, assume responsibility, and gradually form the orchestra's own artistic aspirations. External professional support should help students improve their judgment and practical ability rather than completely replace their own thinking, choices, and responsibilities. A student orchestra may accept guidance, but it should not thereby lose its agency. Even when student

conductors are not yet technically mature, the process of students personally assuming artistic work has irreplaceable educational significance, provided that the orchestra establishes reasonable mechanisms for training, supervision, and feedback. A student orchestra is not a simplified copy of a professional orchestra, but a community that combines artistic practice, organizational autonomy, and the cultivation of talent. This text is written precisely to provide relatively reliable methodological support for such independence, so that students can avoid unnecessary mistakes while retaining the possibility of active experimentation and self-directed growth.

I am often told by the people educating me that when later generations perform poorly, the problem is not necessarily theirs alone. Quite often, we failed to guide them well, failed to organize our experience clearly, and failed to give them a sufficiently reliable starting point. In an orchestra with a complicated institutional history such as ours, many student conductors can only take over their duties in haste, bearing responsibility for rehearsals and performances while repeating mistakes for which earlier generations have already paid a price. If one generation leaves only evaluations but no methods, offers no self-reflection, demands that successors mature quickly, yet neglects proper training and handover, then this rupture itself deserves reflection. This text attempts to transform scattered experience, as far as possible, into writing that can be understood, tested, and continuously revised, so that later conductors need not begin again from exactly the same difficulties. It will probably not produce excellent conductors directly, but it should at least help the next generation avoid some preventable detours and go farther on the foundations left by their predecessors.

The author would especially like to thank Zhao Lihao for his achievements in the understanding of Baroque music and philosophy; Zhang Hongrui for guiding me toward a deeper understanding of opera and the broader art of music; Professor Wang Jian and senior student Liu Yifei for discussions of artistic expression in piano performance; Liu Hangyu, Zhu Tianze, Jiang Xulang, Tang Yanyu, and others for their discussions of the current state of orchestral rehearsals; and all friends in the USTC Orchestra, Choir, and Chinese Orchestra for their illuminating conversations.

2 Preface

The first edition of this guide was written at the end of the author's first semester of senior year, when he served as president of the Student Orchestra of the University of Science and Technology of China. Upon entering the university, the author joined the former Student Symphony Orchestra and the Student Philharmonic Association. At first, he had only amateur experience in piano performance. During his first semester, he encountered a simple piano concerto, and from then on began taking part in orchestral rehearsals. Later, because of the needs of the stu-

dent organization, he rapidly began studying conducting theory at the end of that semester, and in the following semester formally began leading or supervising rehearsals. One year later, the author also briefly participated in rehearsals of the Student Chinese Orchestra. In this process, he met a series of senior students who were respected and capable within the major student music-performance organizations and who cared deeply about musical interpretation, and he accumulated substantial experience related to orchestral rehearsal.

In recent years, all core managers involved in developing the orchestra have found the selection and training of conductors extremely difficult. Historically, the most successful conductors in these organizations have often appeared by chance rather than through deliberate cultivation. Moreover, they have not performed particularly well in training and selecting successors, and have sometimes left the organization with new problems. At the same time, amateur orchestras are inherently unstable and their technical levels vary considerably, which makes their circumstances fundamentally different from those of professional orchestras. Consequently, amateur conductors face situations and require strategies that are not entirely the same as those of professional conductors. In any case, a good conductor can strengthen the cohesion of an orchestra as a student organization and advance its artistic development, whereas a poor conductor can easily cause members to leave and destroy the foundations established by previous generations.

During his brief orchestral career, the author came to appreciate both the difficulty of training and selecting amateur conductors and the orchestra's troubled history. Through conversations with predecessors, he accumulated a number of relevant lessons, but for a long time lacked both the awareness and opportunity to summarize them systematically. Some time ago, while organizing a tutorial left by earlier members on how to hold concerts, the author suddenly conceived the idea of summarizing experience in conducting and leading rehearsals for the training of potential future conductors. In practice, however, this proved to be a formidable task. Conducting remains a discipline without a complete theoretical foundation. It is profound and extremely fragmented, and the required qualities intersect in complicated ways, making them difficult to summarize clearly and logically. The author has made a simple and rough classification, attempting to be comprehensive while simplifying details. The purpose is to give a newly appointed candidate a concise understanding of the matters requiring attention when conducting an amateur orchestra, and to provide a memorandum that can be consulted during training.

The author believes that writing such a guide is pioneering within China. In Chinese-language magazines, journals, and books, a tutorial written in such detail specifically for the training of amateur-orchestra conductors is almost unheard of. The author also hopes that more young colleagues will use this guide, improve and supplement its recommendations and conclusions through practice, and add newer and more effective insights in future versions. Training along the path specified in this guide cannot guarantee a sufficiently high level of conducting,

but it can help ensure the orchestra's normal operation when the artistic level of its leader is not yet high.

The author would especially like to thank Liu Yifei and Zhang Hongrui for their encouragement and advice. Without their continuing exchanges with and instruction of me in music theory and musical practice, this guide might never have been written.

Conducting is a vast art, and many aspects relevant to the conductor of an amateur orchestra remain unmentioned here, including repertoire selection, student-organization management, and artistic administration. The author apologizes for every omission and oversimplification in this text and hopes to improve them gradually in future editions. At the same time, all readers should continue to place musical practice first. Practice is the sole criterion of truth; we advocate active thought and flexible responses to different circumstances.

3 The Composition of an Amateur Symphony Orchestra

A (Western) symphony orchestra consists of performers playing different types of Western instruments, and its size may vary considerably.

Its sections include:

1. String section: violins (divided into two groups), violas, cellos, and double basses. Viola players can be trained relatively quickly from a group of violinists with solid fundamentals, although they must overcome a variety of difficulties, such as reading the alto clef, and may still be unable to master difficult textures quickly.
2. Woodwind section: flutes, clarinets, oboes, bassoons, and related instruments. Oboes and bassoons are generally very scarce. The former are often replaced by soprano saxophones, and the latter by an electronic keyboard or a small group of low strings. When the original part carries an independent melody, the range, timbre, and dynamic capacity of the substitute instrument must be considered.
3. Brass section: horns, trumpets, and low brass. Because horns are particularly scarce, alto saxophones are often used as substitutes.
4. Percussion section: timpani, snare drum, and, where available, bass drum, tambourine, triangle, xylophone, glockenspiel, gong, and so forth. The first two are the most important.

For an amateur orchestra, limitations caused by unstable personnel—which may differ greatly even between adjacent semesters—create persistent additional difficulties in acoustic balance and overall timbral control. More importantly, recruited players differ greatly in orchestral experience. Some students have already spent years in secondary-school orchestras, whereas many others have never participated in a large ensemble. This substantially increases the demands

placed on the conductor.

4 Who Is Suited to Conduct an Amateur Symphony Orchestra?

Before beginning the discussion, I must offer a warning: conducting is, in general, a demanding task. The professional threshold for conducting an amateur orchestra is not as high as commonly imagined, but the overall requirements for basic musicianship and general personal capability are instead quite substantial. If you believe that a conductor only needs to beat the correct time, why do we need a conductor at all?

First, an orchestral conductor should ideally have mastered one instrument, or at least possess broad knowledge and considered views concerning instrumental performance. In the context of an amateur orchestra, “mastery” may mean being able to give a reasonably polished complete performance of a substantial work on one instrument.

Historically, most conductors have come from string playing or piano, but this does not mean that only these musicians can study conducting. In an amateur orchestra, I still recommend that strong string players serve as section leaders, because a capable string player is equally indispensable, and playing one’s own instrument is often easier or more enjoyable than leading rehearsals. Naturally, when personnel are scarce, a talented concertmaster or principal string player often has the ability to take over conducting duties quickly.

For pianists, however, the special nature of the instrument means that their early training is often disconnected from the orchestra. Their advantages are that they can generally read a full score quickly and tend to have good aural and theoretical foundations. Yet conducting an orchestra is much more complicated and fundamentally different from practicing the piano alone. The awareness of musical expression required in orchestral conducting is deeper than that required in piano performance considered in isolation. They are, in fact, two distinct fields. My advice to pianists who wish to conduct is therefore to remain humble, spend more time with the orchestra, learn about a wide variety of instruments from different families, speak frequently with orchestral players, and observe rehearsals. In brief, understand as comprehensively as possible the principles underlying the process of orchestral performance. If such **enthusiasm** drives you to learn quickly, I believe you are not far from becoming a competent or even excellent conductor. At the same time, one should recognize that the source of artistic expression in piano performance and in orchestral performance is fundamentally connected. If one can perceive clearly the horizontal and vertical **structure** of music, then, setting technical issues aside, the intellectual difficulty of performing a four-part piano-sonata movement well should not be

fundamentally very different from that of performing a small symphonic movement. This view will be mentioned again in the later discussion of conducting technique.

The most essential quality required of a successful conductor is intense **enthusiasm** for the activity of orchestral rehearsal; this is always the most important factor. (*As time passes, I increasingly feel that this observation is not limited to conducting.*) It includes enthusiasm for studying scores, for interpersonal interaction within the orchestra and its rehearsals, for learning about different instruments, and for music itself, usually communicated through musical expression. A conductor must show wholehearted involvement during rehearsal. Someone who is attracted only to the form or status of conducting while neglecting the commitment and effort required to achieve its purposes is entirely unsuited to the role; this will ultimately cause great difficulty for both the orchestra and the individual. Naturally, where enthusiasm is equal, musical understanding becomes the more important measure.

At the same time, one cannot ignore that at USTC, a university with an exceptionally strong academic and research culture, student-organization work is always secondary, and the overwhelming majority of students must prioritize their everyday studies. For those who pursue academic competition with extreme severity simply for its own sake (*in colloquial terms, competing merely to compete*), and who regard conducting as a waste of time while lacking the corresponding preparation, this is probably not a particularly suitable undertaking. On the other hand, for students whose studies are so difficult that basic graduation is in jeopardy, or who face suspension or withdrawal, we likewise recommend that academics be secured first, even if they possess extraordinary talent for conducting. It must be acknowledged that today very few people can reach a professional conducting career through this route, but that is another question. (*As an aside, historically, successful conductors in the Orchestra, Symphony Orchestra, and Philharmonic Association have all balanced academics and student-organization work relatively well. It seems that ambitious people tend to possess the ability to manage and reconcile the different demands on their time.*)

5 How Does One Learn to Conduct? Technical Aspects

The less responsible but eternally correct answer is: “practice makes perfect.”

If one discusses this question with professional conductors, one may encounter an extremely self-assured assessment: “Conducting cannot be learned. Either you were born to conduct, or you should not study it.” There is some truth in this. Even today, there is not yet a unified theoretical system for the practice of conducting. Published books on conducting contain comments on practice, disputes over viewpoints, and recommendations for performing particular works. Above all, they present formulaic diagrams of standard conducting gestures

and explain how to beat time. Yet how one should actually conduct, and how one should teach oneself to conduct, has nowhere been discussed in complete detail.

5.1 Instrumentation

First consider the strings. The basic principles of the different string instruments are similar. A conductor should know their approximate ranges and be thoroughly familiar with their basic techniques and resulting sounds—to the point of being able to imagine them mentally. In addition, the student should understand how at least one wind instrument produces sound and know its principal technical difficulties, or otherwise have good opportunities to communicate with at least several wind players. Ideally, the student should also find time to handle different percussion instruments. A competent conductor likewise needs a refined understanding of how several basic drums produce sound.

A conductor needs to know the best and most appropriate sound of every instrument in order to express artistic ideas accurately. In an amateur orchestra, of course, there is always a gap between knowledge and reality. It is common for some players to have problems even with intonation, let alone timbral control. Nevertheless, knowing the ideal sound provides a direction toward which rehearsal can work. At the same time, a conductor must recognize reality, compromise with objective conditions when appropriate, and balance ideals against the actual capacities of an amateur orchestra. This point will be emphasized repeatedly below.

5.2 Music Theory

The conductor of an amateur orchestra does not need complete mastery of every detail of professional music theory. Systematically studying a conservatory curriculum is time-consuming and relatively inconvenient to undertake independently. In practice, musical form and orchestration are comparatively accessible: one can obtain a general understanding from books and deepen it through practice. If the orchestra does not rehearse structurally complex art music, the conductor may not need any knowledge of counterpoint. Harmony is much more difficult, but it is not necessary to grind through a classic textbook from beginning to end. For an amateur ensemble, the conductor primarily needs a good aural foundation, an understanding of the basic harmonic patterns in the works being rehearsed, and the ability to locate dissonant or incorrect notes accurately within the full ensemble. What matters is not a particular *harmony in isolation*, but the *process of harmonic motion*, which can assist the analysis of a work.

This does not mean that music theory is useless. On the contrary, without theoretical guidance, how could one produce expressive motion, contrast, and repose? To perform a work on any instrument at a high level necessarily requires analysis of its basic elements; leading an orches-

tra to a successful performance requires even more of this work. The motivation for studying theory may therefore arise mainly from practical need. As a conductor develops, the student also constructs an internal understanding of music theory, and theory built through practice is often more deeply internalized than words learned from a book. *A perhaps similar analogy is that research in most areas of mathematics does not require one first to study every mathematical proof and its foundations, yet a good mathematician deepens an understanding of general mathematical principles along the path of research.*

5.3 Ear Training

Just as childhood piano study often includes sight-singing and ear training, a conductor must be able to identify sounds accurately. This does not mean achieving machine-like scores in a pitch-recognition contest. The student should also learn, as far as possible, to separate pitch from the interference caused by an instrument's timbre and overtone spectrum, particularly among wind instruments in their normal registers. On this foundation, one can practice with melody: sight-read or inspect a melody, motive, phrase, or any musical passage, and analyze the way in which the melody or other basic elements evolve. Where are its points of emphasis, propulsion, and transition? Imagine that you yourself must perform it: how would you make every effort to heighten its expression? Then add basic harmony, practice following harmonic changes beneath the melody, and consider the role of each part within the harmonic texture. The purpose of ear training is not merely to identify a static harmony at every isolated moment, but to hear how different elements are interpreted and transformed within a continuously changing musical texture, thereby developing a clearer mental conception of artistic expression. From this perspective, a four-part piano sonata and sonata form in a symphony are fundamentally connected.

Rhythmic training is another point that cannot be ignored. Within a fixed duration—for example, one measure lasting as long as a metronome pulse of 48 beats per minute—try dividing the measure into two beats, three beats, and so on up to nine, then return through eight and the lower numbers to two beats and finally one. Early attempts are often embarrassingly poor, and the exercise may remain difficult for a long time, but it has a gradual and lasting effect. Encounter different rhythmic patterns until rhythmic transformation becomes as familiar as drinking water. In rhythmically complex music, from Baroque dances to modern anime-related music, the conductor bears the central responsibility for controlling tempo, and these patterns are often less regular than those in traditional repertoire. When working with such music, listen repeatedly to the original pieces in order to gain familiarity with the relevant rhythmic patterns.

5.4 Score Reading and Analysis; Evaluating Repertoire

Reading and analyzing the full score can never be avoided. It is not the same as attending to a single instrumental part. While observing details, the conductor must also attend globally to information about sections and phrases, especially changes in texture and harmony. Correspondingly, one should undertake appropriate analysis of form, harmony, orchestration, and, where relevant, counterpoint. Individual parts cannot be ignored completely either; at important transitions, attention may need to focus on several particular instruments. Notice every detail you can find. Do not merely identify the notes; know or imagine the required dynamics and articulation. Form a complete conception of the pattern of musical development and understand the desired sonic result. This clarifies the conductor's goals and substantially improves rehearsal efficiency. For an amateur orchestra, the conductor must also judge the relative importance of musical effects, always pursuing the essential goals first and refraining from excessive demands concerning secondary goals when time is insufficient. In general, objectives affecting the whole should take priority over those affecting only a local detail.

Score preparation cannot stop at "having looked through all the notes once." The conductor needs to transform the score into information that can guide rehearsal decisions: how the work unfolds; which parts carry the main material; where the ensemble is most likely to lose coordination; which problems must be solved in full rehearsal; and which should be assigned to sectional or individual practice. Familiarity with the score does not necessarily mean being able to recite every note from beginning to end. Rather, without searching with the eyes, the conductor should know where the music is going next, know when important parts enter, and be able to locate quickly the source of problems heard in rehearsal.

Professional training generally requires a conductor to reduce and play the score at the piano, beginning with Bach chorales, then progressing to string quartets and chamber music, and finally to orchestral works. This is excellent training, but it is excessively demanding for amateur students. In any case, before rehearsal, the conductor must know the score thoroughly, memorize as much as possible, and ideally learn it by heart. The score present in rehearsal should function mainly as a prompt; before turning a page or looking down, the conductor should already know the broad direction of the music to come.

Habits of marking a score differ from person to person. The conductor must also confirm the edition with the music librarian so that the full score and parts are not inconsistent. The conductor and librarian should verify that the score and all parts come from the same version, and that measure numbers, rehearsal marks, cuts, repeats, and tempo indications agree. Arrangements are particularly prone to inconsistency between score and parts, transposition errors, or unsynchronized last-minute changes. The conductor should then record the work's basic infor-

mation, including instrumentation, transposing instruments, clefs, meter changes, key changes, tempo indications, and repeats. For unfamiliar transposing instruments, sounding pitch must be confirmed before rehearsal rather than calculated improvisationally on the spot.

Before rehearsal, list the measures that are likely to cause trouble rather than merely becoming familiar with the work from beginning to end. Common risks include complex rhythms, tempo transitions, sudden soft entries, difficult registers, wide leaps, sustained physical demands, rhythmic displacement between parts, errors in the score, and impractical arrangements. This work should begin during the process of deciding whether to select the piece. On this basis, write a brief and concrete plan before every rehearsal. The plan should at least include the principal goals for the session, the passages to be prioritized, the estimated time to be spent, and the minimum standard of completion, with locations and timing specified as precisely as possible.

Before deciding to rehearse a work, the conductor and orchestra managers should at least answer the following questions. Does the present instrumentation basically satisfy the work's requirements, and are the essential parts staffed? Is the individual technical difficulty appropriate for the current players? Is the ensemble difficulty manageable, considering complex rhythmic patterns, frequent tempo changes, a high concentration of free rhythm, large-scale counterpoint, frequent imitation, and complicated textures? Is sufficient rehearsal time available? Does the work offer clear training value for the orchestra's ability or style? Are the score and parts reliable? Does the work serve a reasonable function in performance, considering duration, physical demands, audience reception, and its relationship to other repertoire?

5.5 Music History, Major Composers, and Important Repertoire

A conductor is not required to master the history of Western music, still less to “command several centuries of artistic flourishing.” Nevertheless, I recommend particular attention to Beethoven. This foundational master established much of the basis for the later orchestra and stands at a historical turning point in many respects. Even today, the underlying logic of the orchestra as a performance institution still rests substantially on musical patterns that he developed and expanded. Anyone thoroughly familiar with his nine symphonies will necessarily understand much about the internal operation of an orchestra. My advice is simple: before conducting Classical repertoire, make time to listen repeatedly to Beethoven's nine symphonies. With even a modest investment of study, one can continue to learn new things from them.

With regard to particular composers and works, I believe that an amateur orchestra should generally avoid Brahms and later Austro-German composers unless a demonstrably manageable piece has been identified. Large orchestral works of the late Romantic and twentieth-century

periods often require extensive instrumentation, complex textures, and a high degree of control over sound, so amateur performers must approach them with particular caution. Yet “caution” does not mean categorical exclusion. Verified simplified editions, excerpts, or suitable arrangements can serve as intermediate training material when the orchestra has the necessary sections. For other composers, the main issue is the difficulty of the individual work. In any event, when deciding to rehearse a composer’s music, investigate the composer, the historical circumstances, artistic style, and performance expectations of the period. At minimum, understand the work’s approximate date, genre, basic style, orchestration, and common performance problems. The purpose is not to lecture at length about music history during rehearsal, but to help the conductor make judgments concerning tempo, dynamics, articulation, and phrasing. Otherwise one may witness the absurdity of rehearsing Ravel without understanding Debussy or French Impressionism, or rehearsing a Schubert symphony without ever encountering his piano works. I support broad training in different musical styles, but diversity of choice must account for the cost of rehearsing such music in an amateur orchestra.

Beyond purely orchestral works, the author has increasingly realized that opera played an indispensable role in the development of Western art music and deserves serious attention from students of conducting. Opera places music within character, language, action, and dramatic conflict. Music no longer unfolds merely as abstract form, but must shape characters, organize time, and advance the plot. Several of Mozart’s comic operas exemplify this particularly well: they embody the spirit of Classicism at its core and may be among the most representative and important works of the Classical period. From this perspective, conflict, transition, suspense, climax, and resolution in later symphonic music can often be understood through dramatic thought. For conductors, the value of studying opera is not limited to learning one genre. Many important conductors spent long periods working in opera houses, and this is not accidental. Even when primarily conducting symphonic repertoire, one can learn from opera how to understand music’s internally generated dramatic force and thereby avoid mechanical performance. Of course, this opens another formidable field of study.

Learning has no end. Those who believe they already possess substantial knowledge of the topics above may continue with world music history, Western art history or the history of civilization, the history of Western philosophy, and more. A great conductor also needs great spiritual strength.

It remains necessary to emphasize that a conductor should have personal taste while fully respecting every form of music. Amateur orchestras in particular should avoid the stereotype that art music stands above everything else. In view of objective limitations and practical difficulties, the significance of musical practice is greater. This does not mean that every work offered to an orchestra deserves praise. Some student-composition exercises have technical de-

fects without corresponding artistic achievement; after proper evaluation, such music may not merit rehearsal.

5.6 Basic Conducting Technique: To Beat Time or Not to Beat Time?

Primary-school music textbooks teach how to read and beat duple, triple, and quadruple meter, and in one sense this is already sufficient. More complex rhythms are essentially formed by addition or subdivision; this is the essence of compound, asymmetrical, and subdivided meter. *Whatever beat pattern is adopted, the first criterion is that players can clearly identify the tempo, beat points, entries, and musical character.* The specific method should be selected according to the circumstances, choosing the clearest solution least likely to be misunderstood.

When beating time strictly, the gesture must be **standardized**. This does not mean that its path must exactly reproduce a diagram in a book, but that every player must be able to understand the beat clearly. In principle, the first beat must move distinctly downward, while the last beat should usually have a clear returning direction; their vertical positions should differ unmistakably. In quadruple meter, the heights of the two middle beats may vary, but they should be clearly distinguishable from the first and last. It is especially important that the string section can understand these gestures. First violins and cellos seated to the conductor's sides often cannot see the complete motion. When confused, they need a clear contrast between upward and downward direction in order to adjust. This is easily overlooked.

The conductor must also **develop independence between the two hands**: one hand beats time while the other gives additional cues without interference. There are useful exercises for this as well. After the patterns are familiar, habitually practice a standard beat pattern with one hand during everyday activities while the other hand performs an unrelated task. If the hands can operate without disturbing each other, control of the beat in actual conducting will improve substantially. In general, once good independence has been developed, the right hand maintains the basic pulse while the left hand indicates sections, dynamics, sustained notes, phrasing, and other important directions of musical expression.

How should an entrance be given? This requires focused study and especially intense attention at the beginning. A basic preparatory process can be divided into three stages. First, confirm the entering instrument or section through the eyes or direction of the body. Second, give a preparatory motion whose timing corresponds to the tempo. Third, guide the production of sound at a **clearly defined** beat point. The duration of the preparation must be consistent with the tempo about to begin. An upbeat does not justify an ambiguous preparation; on the contrary, it demands greater clarity of duration and direction. A beginner may precede the entrance with one complete measure as a temporary substitute, but should not rely on this method indefinitely.

A cutoff likewise requires preparation. The conductor must first make clear how long the final note is to be sustained, and then give an unmistakable moment of release—the ability to “hold the sound.” During a fermata or sustained note, the holding gesture should remain stable rather than continuing to beat mechanically. As the end approaches, a slight preparation may precede a clear and unified release. In addition, the end of a phrase is not identical to the end of a work. An internal closure may require only a reduction of gesture or a change in the sense of breath, whereas the final ending of a work normally requires a more explicit release and stillness.

Sectional cues should always be prepared before the moment of entry, generally through eye contact, a turn of the body, or a small redirection of the arm. An effective cue should also tell players how to enter, which depends on the skill of the second hand. An experienced conductor can make this process highly expressive. The same cue toward a flute can indicate a gentle entrance, a firm accent, the beginning of a singing melody, or the need to preserve rhythmic stability within a complex texture. In all cases, the conductor should know the important entrances before rehearsal and, as far as possible, look up to confirm the relevant section in advance.

Tempo is determined first by the time between adjacent beat points, not merely by the size of the gesture. Fast music generally requires compact movements to reduce unnecessary motion. Slow music may use wider space, but must still preserve clear beat points and a continuous sense of time. Players also interpret gesture size as dynamic information, so one cannot simply equate “large movement” with “slow tempo.” Fast and powerful music may require compact but weighty gestures, while slow and soft music may require broad but gentle ones. The conductor must distinguish the range, speed, weight, and clarity of a gesture.

Control of tempo likewise requires long-term practice. Professional conductors can estimate a minute accurately even under distraction. I tend to recommend memorizing a representative work for several common rhythmic characters, thereby developing thoroughly familiar internal tempo models. When a similar pattern appears later, the model can be recalled to help control the overall tempo. Stable tempo production may also require practice with a metronome.

Finally, the effectiveness of a gesture can always be assessed through the orchestra’s reaction. If several sections repeatedly enter early, lag behind, or misunderstand the same location, the conductor should first consider whether the gesture itself is defective rather than immediately concluding that all the players failed to watch.

6 How to Lead a Rehearsal: Communication

Generally speaking, a conductor has three means of expression available: gesture, facial expression, and language. The most important is an expressive and compelling physical gesture. For an amateur orchestra, however, highly refined gestures may not be readily understood, so

language becomes an important auxiliary tool. Language and facial expression can facilitate communication, but they can also hinder it.

Clarity of movement is the first requirement. In principle, the conductor should work mainly with the arm below the shoulder, although the range of motion may sometimes be enlarged moderately to give players a more obvious reminder. In an amateur orchestra, standard gestures and changes of gesture should be made with very clear, sometimes even exaggerated amplitude so that all members can fully understand the intention. For example, in a *ritardando*, the standard beat pattern should expand conspicuously while the motion itself remains as temporally controlled as possible. The conductor then holds the tempo firmly until it stabilizes or reaches the note at the end of the phrase, and releases it on the next attack. An *accelerando* proceeds in the opposite way. More generally, in addition to verbal emphasis, enlarge gestures as needed to maximize their signaling function or expressive force.

Different musical materials and articulations require different gestures. Long, continuous, singing passages need gestures with a singing quality, as if producing waves in a viscous liquid, without abrupt beat points. Passages that emphasize granularity, marked staccato, or frequent breaths require a steadier transmission through the same imaginary medium, with composure, precise control, and clear beginnings and endings.

Every artistic conception should have a corresponding movement. The conductor's musical idea should generate the gesture, rather than technical habit determining the imagined sound. In principle, the conductor should embody the entire conception in physical language. This is an extremely demanding standard, and failure to reach it fully is understandable. At the same time, an amateur orchestra has limited ability to interpret obscure gestures. A balance must therefore be found between speech and gesture, seeking the most efficient mode possible. It is usually best to align rhythm and intonation first—which means working through the entire passage with a clear standard beat—and only then pursue higher artistic goals. *One potentially efficient method is first to provide mechanical beat points in the simplest and most direct way, through tapping or a metronome-like pattern, so that the ensemble can align at a basic level.*

Yet the ideal state of instrumental performance is to express the work's process of development, a process that should arise naturally from the work's own forces rather than being artificially imposed. A difficult balance therefore exists between conducting and overconducting. Even many experienced professional conductors are sometimes uncertain about this. It is enough for the student conductor to remember the principle and apply it when appropriate. A less mentally taxing method is to spend as much attention as possible on global matters beyond mere accuracy.

When should one beat time? At the beginning of rehearsing a work, I generally recommend starting with a strict, regular beat pattern unless the piece has unusual rhythmic requirements.

Being able to beat time correctly is sufficient for more than ninety percent of repertoire. At the same time, the conductor's ultimate aspiration should not be to remain a human metronome almost all the time. Within the balance described above, I recommend giving phrasing suggestions through physical language whenever possible. Even while the right hand maintains a standard beat, the left hand can offer more expressive indications.

How should one choose an interpretation, and how should arrangements be regarded? For an amateur orchestra, an artistically less orthodox solution may be acceptable. Appropriate changes in orchestration or tempo, and further interpretation of the written text, are entirely permissible. Musical practice is always the most important consideration. An amateur orchestra should adopt a more pragmatic perspective, seeking to complete the practical process at the highest possible quality rather than following an almost fundamentalist doctrine. Naturally, I also do not support excessive alteration of famous canonical works, but under most conditions those works are likewise unsuitable for an amateur orchestra.

7 Orchestral Training: Rehearsal Technique

In principle, the most important consideration in a large-ensemble rehearsal is **efficiency**. What people call a conductor's ability to lead rehearsal mainly consists of the ability to **identify problems, break them down, and solve them**.

For an orchestra, the conductor's greatest function is to **save time**. Accordingly, when evaluating a conductor, one often focuses on the amount of rehearsal time used effectively. During preparation, the conductor must possess a highly concrete conception of every note. Only then does rehearsal have a clear direction, and only then can the players be persuaded by the conductor's work.

Remember firmly that rehearsal primarily serves the main musical elements: **intonation and relationships among parts, thereby establishing basic ensemble; tempo and rhythm; dynamics; and articulation**. More refined phrasing and stylistic interpretation come afterward. The first three should be central priorities for an amateur orchestra. Quite often, if suitable intonation and dynamics are firmly established and the performance proceeds at a reasonable tempo with good coordination, the resulting performance is already very effective.

First, formulate a **strict rehearsal schedule**. In other words, after spending some time with the orchestra, suppose thirty minutes are available to rehearse a five-minute work: how should the conductor allocate them? The plan need not be perfectly precise, but the conductor should know clearly what is to be done. For an amateur orchestra, it is especially important to balance the conductor's ideal vision against the ensemble's actual ability. The ideal should remain adjustable. By continually assessing the orchestra's condition and determining what is genuinely

needed, the conductor can identify more clearly what must be emphasized in rehearsal. Do not spend excessive time on a single special passage, and do not insist on rehearsing from beginning to end. The central function of rehearsal is to solve problems and work on passages that require coordination. Except during the final two or three sessions in a planned sequence, there is no need to demand a complete run-through of the work.

Second, *do not lecture excessively* during rehearsal, and do not overemphasize theoretical analysis. Performers often care less about theory and historical stories than the conductor expects; some may become irritated by or suspicious of narrative descriptions. Appropriate rhetoric can be useful, but it should be controlled and should serve only to help players understand the artistic conception. This is a balance between maintaining an engaging atmosphere and rehearsing efficiently. Precise language that helps realize the desired technical and artistic result should, by contrast, be used fully and is rarely excessive; it also plays an important role in mutual understanding between conductor and players. For example, while rehearsing the *Egmont* Overture, retelling the story of the Dutch revolt is far less useful than telling the string players how to execute the tremolo as the motive develops and how much vibrato to use. The players may then explain that the notes are too high for reliable vibrato, after which the conductor considers the matter carefully and abandons that demand. This can be an efficient exchange for both sides.

When should the playing be stopped to correct an error? As a general rule, at the beginning of rehearsing a piece it is best to proceed through most of it without interruption, restarting only when the ensemble has fallen apart so badly that coordination is impossible. During later refinement, an intonation problem caused by an individual player's temporary condition or mistake, or a minor rhythmic error, may be ignored. A deficiency of expression can often be addressed verbally while the music continues. Likewise, *there is no need during full rehearsal to drill a single section repeatedly or make each player in the section perform the same passage one by one*. Problems confined to a section's own technique or caused simply by insufficient practice should, once the section understands the musical development, be resolved in later sectional work rather than consuming the entire ensemble's time. Similarly, excessive fault-finding over small defects can cause misunderstanding or resentment. Detailed problems concerning a particular player, long solo passages, or recommendations about personal playing style should be discussed privately, with their treatment during full rehearsal reduced as much as possible. One further point is essential: after stopping, always explain why. Do not say nothing and merely demand "again," unless the ensemble has already collapsed so completely that the reason is self-evident.

How should one communicate concretely with players of different instruments? Express the request through specific technical advice with which you are familiar. If a special effect

is required, sing it, however poorly, rather than relying only on verbal description. Also make sure that the players can follow your location. When giving detailed technical suggestions, for example, identify the relevant measure number. Merely referring to the development or recapitulation may not allow most players to keep pace.

When a difficult passage never comes together, remember how you practice your own instrument. A common approach is to control variables, beginning from the simplest possible state and adding complexity. For piano, the usual methods are slowing down, separating the hands, and repeating. Orchestral rehearsal is similar. One may need to reduce the tempo, retain only several parts, slow a local passage, or even tap out the beat one pulse at a time—which is often very useful. Then add instrumental groups layer by layer, repeat the passage, and gradually accelerate. Naturally, if repeated attempts continue to fail, reconsider promptly whether the repertoire presents an inappropriate technical difficulty, and try to prevent such a situation during selection and preparation.

What is the relationship between beating time and changing tempo or dynamics? In principle, tempo influences the scale of movement: faster music tends toward smaller gestures and slower music toward larger ones. The conductor should control an even transformation of space and time, meaning that the internal motion of the beat remains stable at different tempi, as discussed earlier. Dynamics are often shown directly in the score. Additional adjustment and emphasis can be communicated through the vertical range of the gesture, eye contact, or indications from the non-beating hand. Players are often less sensitive to tempo changes, however, so those usually require stronger physical emphasis.

8 Orchestral Training: Psychology

Many anecdotes describe the mysterious world of orchestral rehearsal. People enjoy books and articles that report the legendary abilities and eccentricities of famous musicians. These stories, however, offer limited practical guidance to younger colleagues. Broadly speaking, a conductor assumes responsibility not only for artistic decisions, but also for organizational communication and rehearsal order. A conductor in a student orchestra usually lacks the absolute authority supplied by a professional institution. Influence therefore comes mainly from thorough preparation, clear expression, reliable judgment, respect for members, and a genuine ability to improve the quality of rehearsal.

The conductor is not merely the artistic leader of the orchestra. In fact, a conductor should combine the qualities of educator, leader, and collaborator. The educator helps players understand the work and improve ensemble; the organizer determines rehearsal goals and allocates time; the collaborator recognizes that performers possess more direct experience of their own

instruments and is willing to consider reasonable opinions. None of these identities can replace the others. There are always people in an orchestra who doubt the need for a conductor and believe that their interpretation is independent of the conductor's presence. A good conductor should be able to go beyond the players' existing technical capacities and guide them toward revealing the best qualities within themselves—that is, to help them discover a musical understanding beyond their present selves. A conductor who achieves this will certainly earn the orchestra's respect, but doing so likewise requires full respect for the performers. This is the process by which authority is established. Here, "rehearsal order" does not mean demanding unconditional obedience. It means creating a working method that all members can anticipate: how problems are raised, how decisions are formed, how disagreements are handled, and what responsibility each person bears.

For every new conductor, the most difficult initial task is undoubtedly to establish authority quickly in rehearsal, or equivalently, to make the orchestra believe that the conductor's preparation and judgment deserve serious attention. Such credibility cannot be obtained solely through title, tone of voice, or the scale of gestures; it must be built gradually through repeated rehearsals. This is difficult because the people being guided are often more experienced than the conductor. Moreover, the process varies greatly by person and circumstance, and every conductor must find an individual way to communicate feeling. Nevertheless, deep knowledge of the score and its musical style, clear physical expression of a musical conception, and strong enthusiasm can overcome much of the orchestra's initial doubt. Credibility comes chiefly from several sources: familiarity with the score and the ability to locate problems quickly; requests that are concrete and executable; acknowledgment of unfamiliar technical questions; prompt correction after a mistaken judgment; and respect for rehearsal time, without repeated and purposeless demands to "do it again." A conductor need not conceal every error. If the wrong measure number is given, the orchestration is misread, or an unreasonable request is made, admit it briefly and correct it immediately. Acknowledging an error usually does not weaken credibility. Repeating the same kind of error while refusing correction is what causes players to lose trust. Authority should not be established by force; the conductor of an amateur orchestra should not be dictatorial or obstinate. Personal judgment is necessary, but self-righteous stubbornness must be avoided.

To gain acceptance of the conductor's authority, the conductor needs to know how to maintain order while showing personal appeal, such as humor or broad knowledge, and must understand both strictness and tolerance. A good conductor enables the orchestra to recognize the importance of collaborative rehearsal and to learn something new from it, thereby sustaining members' interest over time. If many people begin missing rehearsals without explanation, or spend most rehearsals with nothing to do, the conductor should consider whether this is partly a consequence of the rehearsal itself.

How should the conductor manage interpersonal relations with the orchestra? Fundamentally, the conductor must know what to say and what not to say. The wording of criticism, the moment for encouragement, and the appropriate occasion for a joke all affect the relationship between conductor and ensemble. The conductor should not always try to please everyone. Major errors of playing habit or musical expression should be identified directly. On the other hand, one should not offer extreme personal judgments about a player's performance casually, unless the relationship is exceptionally close. In musical matters, the more specific the feedback, the easier it is to execute. Vague, impressionistic comments such as "make it better," "give it more feeling," or "do not be nervous" usually do not solve a problem effectively. The conductor should criticize what happened in the playing rather than judge the player as a person. The general principle remains that the conductor should show respect for every performer, both artistically and interpersonally. Only then can respect be returned and members devote themselves fully to rehearsal. There is no need to attach mechanical praise before and after every correction, but players should know whether their effort has produced a result. A long-term pattern of negation without feedback about improvement will reduce commitment.

Full rehearsal is an appropriate place to address common rhythm, sectional balance, unified articulation, harmonic relationships, and the structure of the work. A problem shared by an entire section may likewise be addressed openly to that section. Individual technical deficiencies, persistent solo problems, personal playing habits, and sensitive matters of attitude are generally better discussed privately after rehearsal. Publicly naming a player is sometimes necessary, for example to confirm that a soloist understands a measure location, but full rehearsal should not become a prolonged public examination of one member. In private conversation, the goal should likewise be to solve the problem. The conductor can first ask what difficulty the player is experiencing, then jointly determine whether individual practice, sectional rehearsal, or an adjustment of the conductor's demand is needed.

The conductor should not bypass the concertmaster and section principals in dealing with every specific technical matter. String bowings, wind breathing, and the internal distribution of work within a section normally require the participation of the relevant leader. When a technical dispute arises, first describe the required musical result and then invite the player to propose a feasible method. The conductor retains responsibility for the overall artistic judgment, but need not pretend to understand every technical detail of every instrument better than the performers do. When a concertmaster or principal offers a different opinion, it should not immediately be regarded as a challenge to the conductor's position. First determine whether the disagreement concerns technical feasibility, sonic aesthetics, or rehearsal organization.

Rehearsal order is best supported by rules stated in advance rather than punishment improvised according to the conductor's mood. Frequent absence may result from personal irre-

sponsibility, but it may also reflect unreasonable rehearsal arrangements, excessive waiting, a long lack of progress in the repertoire, or members' uncertainty about their role. When absence becomes widespread, managers must examine both the organizational system and the quality of rehearsal rather than attributing every problem simply to attitude. Basic quiet is necessary during rehearsal, but a student orchestra should preserve some room for communication and a relaxed atmosphere, provided that this does not continually interrupt the work or prevent others from hearing instructions.

The conductor's emotional state rapidly influences the whole orchestra. Commitment and enthusiasm can raise attention, but agitation, disappointment, and anger are also magnified. The conductor needs to distinguish between "the work has not reached its goal" and "the members do not respect me," and avoid transforming musical problems into personal conflict. When a passage repeatedly fails, change the method before merely raising the voice or repeating the same demand. Slow down, reduce the number of parts, alter the explanation, take a brief rest, or move temporarily to another passage. Repeatedly using the same ineffective method only accumulates frustration for everyone. After rehearsal, the conductor should review the session to an appropriate degree. Growth comes not only from studying the score, but also from continually correcting one's rehearsal behavior.

Because we are a student organization, we do not engage in matters such as communication with sponsors, and those topics will not be discussed here.

9 Instrumentation for Amateur Orchestras

The following subsections do not constitute a complete course in instrumentation or orchestration. They summarize only the problems most commonly encountered in rehearsals of amateur orchestras. Instrument models, the technical habits of different performers, and different musical styles vary greatly. This text can therefore provide only directions for judgment; it cannot replace communication with the actual players.

A conductor does not need to be able to play every instrument, still less master every branch of instrumentation. The conductor does need to know whether a demand is technically possible, how much practice it will cost, how much difficulty it creates for the player, and what sonic result it may produce. When encountering an unfamiliar technique, ask the principal or an experienced performer before issuing a demand based solely on terminology in the score.

The ranges, techniques, and timbral tendencies mentioned in this chapter should be understood as general cases. For a particular work, the final judgment must be based on the written score, the actual instrument, and the player's present condition.

9.1 General Considerations for String Technique

An accent produced by the finger differs from one produced by the bow. The latter may support a crescendo, while the former emphasizes a particular note and increases clarity. The appropriate intensity of vibrato must be considered carefully for different works.

Changes of bow direction have natural tendencies: an up-bow tends toward crescendo and a down-bow toward diminuendo. The up-bow can be understood as inhaling and the down-bow as exhaling. Bowing should be used correctly and with awareness of the special implications of full bow, half bow, and one-third bow, as well as the middle, frog, and tip. Encourage all players to use the bow fully. Bow changes within a continuous phrase must be smooth and natural, without pauses or unintended accents. In highly emphatic music, unified points of bow change may help create a common accent.

For separate bows, ordinary staccato, spiccato, and pizzicato, the conductor should understand the detailed differences in timbre created by each technique and provide appropriate guidance whenever possible.

Other, more modern string techniques can generally be ignored unless a special piece requires them.

The viola section should receive additional attention during rehearsal because severe imbalances in the orchestral sound can make its errors difficult to detect. This observation does not apply when the section is sufficiently large.

9.2 General Considerations for Woodwind Technique

Always attend to the intonation of the woodwinds. Even after careful tuning, different instruments may deviate considerably in their high and low registers. The presence of a soprano saxophone makes this problem more difficult. It is also necessary to understand that different woodwind instruments have different levels of difficulty in controlling dynamics across registers. For flutes and recorders, for example, playing softly in the upper register and strongly in the middle and lower registers can be very difficult. The capabilities of the different woodwinds and their acoustic balance therefore form a substantial field of study. Orchestration may require many adjustments, and when a composer writes a difficult passage the conductor should understand adequately the challenge facing the players.

Breath management and the placement of breaths are especially important for woodwinds. A clear attack or forceful entrance requires a different method of blowing from an indistinct attack or gentle entrance. The former uses a rapid release of the tongue, while the latter requires prepared air to be released more gradually from the mouth.

Breathing points are usually directly related to phrase structure. The conductor should

allow performers to breathe where necessary and discuss staggered breathing with the section principal when appropriate.

The effect of rapid staccato or tonguing depends on the player's technique and current condition. If a performer cannot meet the demand reliably, respond with understanding. For a rapid figure that cannot be stabilized, try lowering the tempo, rephrasing it, or redistributing the part rather than repeatedly requiring the performer to demonstrate technical ability in front of the full orchestra. Woodwinds possess a rich range of expressive colors. Further study of orchestration and instrumentation is needed to understand the characteristics of individual woodwinds and their combinations.

9.3 General Considerations for Brass Technique

Our orchestra has a long tradition of diligent brass practice, so the players' knowledge of specific techniques will certainly exceed that of a conductor without specialist training. Frequent communication with them can resolve many questions.

For brass players, the ability to sustain correct notes steadily already satisfies many requirements. If dynamics can also be controlled correctly and reasonably, the result is quite good. Compared with strings and woodwinds, sustained brass playing consumes more physical strength and is more dependent on the player's condition, a fact the conductor must understand during rehearsal. Because inhalation, embouchure preparation, and sound production all take time, brass entrances have a certain inherent delay. Appropriate advance cues should therefore be given after long rests.

When communicating with brass players, specify the attack, note length, upper dynamic limit, and treatment of the phrase ending. When the conductor does not know the exact playing method, describe the required sound and allow the performer to propose a workable solution.

Brass balance cannot be solved merely by asking players to "play more softly." Horns sometimes serve as inner voices with a woodwind-like function and at other times join the brass tutti. Trombones and tuba may provide the harmonic foundation or create powerful rhythm and accent. The conductor should first identify the orchestral function of the brass in a particular passage before deciding whether it should project or blend.

9.4 General Considerations for Percussion Technique

For the percussion section, correct playing technique is even more decisive than in other sections because it directly determines sound quality. Percussion parts may contain relatively few notes, but those notes often carry important structural, temporal, climactic, or timbral functions. Few notes do not imply low technical demands. Striking position, choice of mallet,

method of attack, roll speed, and the timing of damping can all alter the overall sound significantly. Timpani and snare drum are highly important in many works, but the central percussion instruments differ from piece to piece. Cymbals, triangle, tambourine, glockenspiel, or gong may likewise carry essential structural functions. When communicating with percussionists, specify the attack, resonance, and cutoff. Rhythmic stability is especially important, particularly when the percussion section establishes a clear underlying pattern.

10 Conclusion

The art of conducting is highly complex. In addition to technical ability, the combined balance of objective reason, personal initiative, emotion, decisiveness, and interpersonal skill is a determining factor in leading an orchestra.

At the highest level, the conductor's activity is connected with the spirit, one of the most powerful human forces and often the strongest form of artistic expression. The conductor's mission is therefore elevated: it requires spiritual strength, knowledge, technique, emotion, and creativity to be joined closely together.

From this perspective, a conductor should also learn more about Beethoven's great spiritual force—his struggle against fate and his endless resistance. The call issued by the solemn double fugue in the Ninth Symphony demands clearly: people throughout the world, unite! Yet his lonely mind, ahead of its time and understood by few, also feels anxiety and asks: can the people of the world truly unite?

This is undoubtedly difficult. The art of conducting is precisely the art of bringing all the people of an orchestra together. As this guide of moderate length has shown, even that task is full of challenges. An orchestral conductor must find a path toward solving the problem. The answer should not rely only on the recommendations given here, but must also be discovered through practice. May the conductors of amateur student orchestras balance an elite path with a broadly participatory one and, through practice, find the best way to lead their ensembles.

11 References and Further Reading

For various reasons, it is difficult to provide a formally typeset bibliography here as one would in a conventional academic paper. Only scattered suggestions for further study can be offered.

All conducting manuals should be inherited critically, and claims inapplicable to amateur orchestras should be discarded. The most useful material in such books is often their analysis of score examples.

Teacher Wang Leyou's video series informally titled *Learn Some Damn Conducting* and *Learn Some Damn Music Theory* is well worth watching and contains valuable material. This text has also been influenced by ideas presented there.

For the four principal areas of music theory and related tutorials, one can search online. A large proportion of Chinese translations of textbooks are obscurely written and do not necessarily preserve the original meaning well. Note that instrumentation is not identical to orchestration; for a conductor, the former is more important.

For musical style and history, several widely recognized books are available, including *The Norton Introduction to Music History* series and *The Classical Style*. In studying the styles of major composers, Beethoven may serve as a central point. Looking backward, one may study Mozart, Haydn, and Bach; looking forward, Schubert, Schumann, and later Brahms, Wagner, Bruckner, and Mahler. For French styles, particular attention can be given to Debussy, Ravel, and Bizet. Italian opera has exerted an important influence on many forms of music. Russian composers may be approached through Tchaikovsky and then Rachmaninoff, Shostakovich, Prokofiev, and others. Nationalist and more modern schools include many other celebrated composers, some of whom combine influences too complex for forced classification. Many important figures remain unmentioned, and learning their works requires substantial time.

Learning has no end. The ability to search effectively, think independently, and learn what is needed when it is needed remains the basic method for entering an unfamiliar field quickly.